

Bryan Zin

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Education

Cornell University

Ithaca, NY

B.S. in Mechanical Engineering

Dec 2021

Relevant Coursework: Linear Algebra, Probability & Statistics, Machine Learning (OCW 6.036), Reinforcement Learning (Stanford CS234)

Research Statement

I'm interested in applying machine learning and computational modeling to understand global manufacturing and industrial systems related to the interdependence between the U.S. and Asia. I'm particularly interested in predictive methods (multi-agent models, reinforcement learning) that study how firms, industries, and production networks respond to technological, economic, and geopolitical change.

Research Experience & Projects

Supply-Chain Risk & Financial Markets

Independent Research

2026 - Present

- Goal: Test whether elevated supply-chain risk is associated with negative abnormal returns.
- Developed Python pipeline to collect/analyze SEC 10-Ks, XBRL, and equity-market data across 450 public companies in 9 sectors.
- Used Word2Vec to create supply chain risk and resolution dictionaries from 20 years of SEC filings/earnings calls as context.
- Applied FinBERT to quantify information contained in corporate disclosures and investigate relationships to market returns.
- Built event-study and regression models comparing actual cumulative abnormal returns (CAR) to expected returns using a 4-factor benchmark, isolating firm-specific market responses from broader market movements.

Machine Learning Reproduction & Experimental Projects

Independent Projects

2026 - Present

- Reproduced optimization experiments from the Adam paper (Kingma & Ba, 2015), implementing Adam, AdaGrad, RMSProp, and SGD/Nesterov and comparing convergence behavior on neural-network (NN) training loss.
- Implemented and evaluated NN architectures using NumPy and PyTorch, including MLP and convolutional models on MNIST/CIFAR-10.
- Designed controlled experiments to investigate optimizer hyperparameters, convergence behavior, dropout, and model architecture.

Industry Experience

Reframe Innovations, Inc.

San Francisco, CA

Founder

Jan 2026 - Present

- Accepted into Y Combinator (W26) and raised \$1.2M in seed funding to build AI-driven procurement automation for hardware teams.
- Architected automated workflows across bill-of-materials management, vendor quoting, POs, and component tracking.
- Engineered data pipelines to centralize vendor purchasing metrics, providing real-time, sourcing visibility for engineering groups.
- Built seamless integrations with core enterprise software ecosystems, encompassing email, Slack, logistics API feeds, and ERP systems.
- Acquired 6 customers (including Eight Sleep, Revise Robotics, Retina Robotics) generating \$285K in annual contract value.
- Managed customer relationships from initial outreach to product demonstrations to terms, negotiations, and signed contracts.

Freshfleet Robotics, Inc.

San Francisco, CA

Founder

Aug 2025 - Nov 2025

- Designed custom end-effectors for autonomous in-vehicle cleaning, integrating electrical systems and circuit design.
- Secured pre-seed venture capital funding (\$130k) and constructed fully functional, prototype units.
- Developed microcontroller firmware to execute low-level actuation and hardware control, achieving 3-min cycle time per seat.
- Implemented real-time path planning and dynamic collision avoidance algorithms via ROS 2 on UR10e industrial manipulators.
- Managed an external software contractor to accelerate robotics programming and optimize motion planning.
- Led deployment negotiations with major fleet operator, Moove, to validate technical specs against operational constraints.

Apple Inc.

Cupertino, CA

Product Design Engineer III

Jan 2022 - May 2025

- Owned mechanical architecture design, structural integration, and rigorous testing for several flagship device generations.

Key contributions across device generations (iPhone 15 Pro Max, iPhone 17 Air, and iPhone 17).

- Formulated validation protocols and evaluated yield datasets to assess critical failure modes and verify engineering modifications.
- Designed the display frame
- Leveraged statistical factory analytics and empirical testing to rectify production bottlenecks, driving >\$50M in yield optimizations.
- Spearheaded international technical investigations and executive presentations with eng/ops teams in US, China, India, and Singapore.

Meta Platforms, Inc.

San Francisco, CA

Product Design Engineer Intern

Jun 2021 - Aug 2021

- Spearheaded physical design for experimental privacy slider mechanism on Oculus Quest 2
- Engineered plastic enclosures incorporating DFM/DFA principles, injection moldability, and detailed tolerance stackup analyses.
- Delivered user research and prototype findings to industrial design groups, enabling a 20% size reduction in privacy switch footprint.
- Conducted ergonomics research across 20+ participants to establish ideal button geometry, travel distance, and accessibility metrics.
- Executed thermal and humidity testing on Oculus Quest 2, capturing over 20,000 telemetry data points for environmental modeling.

Barclays PLC

New York, NY

Summer Analyst

Jun 2020 - Aug 2020

- Evaluated credit line reduction strategies across Barclaycard portfolios using underlying model analytics, regulatory standards, and commercial exclusion policies.
- Performed driver analysis to determine key operational and policy variance across US, Germany, UK, and Small Business card divisions.
- Refined targeting parameters for US credit reduction frameworks, reducing monthly net credit losses by \$13.4M in July 2020.

BAE Systems, PLC

Nashua, NH

Mechanical Engineering Intern

Aug 2019 - Dec 2019

- Architected hardware interface integrating APKWS electronics with Honeywell IMU, utilizing Creo Sheetmetal for flex circuit layout.
- Executed ANSYS finite-element structural and buckling simulations across diverse geometric configurations
- Generated STAR-CCM+ computational fluid dynamics models evaluating missile aerodynamic profiles to reduce unit production costs.

Skills

Programming: Python (PyTorch, NumPy, Pandas, Scikit), Matlab

Machine Learning: NLP, transformers, neural networks, reinforcement learning, optimizer tuning, supervised learning, PPO, DPO

Awards

US Patent #20240064220A1: Handheld Electronic Device — Patent for a handheld electronic device focused on mechanical architecture design and structural integration of the display module into metal housing.

Publications & Manuscripts

Zin, B. "Supply-Chain Risk Disclosures and Equity Market Abnormal Returns." (Working Paper, 2026).